

The impact of daytime melanopic light dose on evening melatonin suppression

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Abstract

Non-visual responses to light, including circadian phase entrainment, melatonin suppression, and pupil response, are mediated by melanopsin-expressing intrinsically photosensitive retinal ganglion cells (ipRGC). Melanopic equivalent daylight illuminance (mEDI) quantifies melanopsin stimulation and is used in recommendations for healthy light exposure. However, responses to standardised light environments vary substantially between individuals for circadian photosensitivity, particularly in melatonin suppression. Although sex, age, genotype, and recent light history may contribute to this variability, evidence concerning recent light history has mainly come from controlled laboratory conditions. It therefore remains unclear whether naturally occurring day-to-day variation in real-world daily light exposure (DLE) is associated with subsequent evening melatonin suppression. Here, we will test the confirmatory hypothesis that, within individuals, daytime mEDI dose preceding an evening laboratory assessment is associated with melatonin suppression. As hypothesis-generating questions, we will also examine whether between-person differences in average daytime mEDI dose and other DLE characteristics are associated with melatonin suppression and melanopsin-mediated pupil response. We will use a repeated-measures protocol combining continuous real-world mEDI measurements with laboratory assessments of melatonin suppression in healthy young adults. This study will provide the first systematic examination of whether naturally occurring differences in DLE predict individual neuroendocrine photosensitivity.

Introduction

Exposure to light regulates various physiological functions in humans, including the production of the hormone melatonin, the circadian timing system, alertness, and pupil size. These neuroendocrine, circadian, cognitive and pupillary effects of light exposure are termed “non-visual effects” and are mediated by the melanopsin-containing intrinsically photosensitive retinal ganglion cells (ipRGCs) in the retina¹. The contribution of melanopsin-containing ipRGCs to steady-state pupil size and melatonin suppression can be appreciated by looking at the spectral sensitivity of these responses, which is consistent with that of melanopsin, with a peak at approximately 480 nm². Modern individuals spend a significant portion of their days indoors, where light levels are substantially lower than those in natural outdoor environments³. In contrast, they are frequently exposed to bright artificial lighting at night. This modern pattern of light exposure is associated with increased health risks, including sleep disorders, depression and diabetes, through disruption of the body clock that regulates sleep/wake cycles⁴.

In recent years, international standards such as melanopic equivalent daylight illuminance (mEDI) have been established to quantify melanopsin stimulation based on the spectral composition of light. Expert recommendations for healthy light exposure, formulated using mEDI, aim to support proper circadian rhythms⁵, and recent research has focused on designing appropriate light environments to achieve these goals. However, standardised light environments may have limited effects on some individuals because there are large individual differences in melatonin suppression, which is widely used as a marker for light responsiveness of the circadian system⁶. In fact, blue light interventions aimed at regulating circadian rhythms can cause some individuals to shift their circadian rhythms by about 3 hours, while others have little or no effect⁷. Therefore, there is a need to clarify the determinants of individual differences in light responsiveness of the circadian system, such as melatonin suppression, to propose healthy light environments for each individual.

Most previous research to determine the factors affecting individual differences in melatonin suppression has focused on relatively stable between-person characteristics, such as sex, age, and genotype^{8,9}. In addition to these demographic and genetic characteristics, recent light exposure may be a potential determinant of melatonin suppression. Controlled laboratory studies have shown that prior exposure to dim versus brighter light conditions can alter subsequent melatonin suppression in the evening^{10–12}. Specifically, melatonin suppression was lower when the same participants had previously been exposed to the brighter condition than when they had been exposed to the dim condition. Together, these studies demonstrate a light-history-dependent effect under controlled laboratory conditions. In everyday life, daily light exposure varies substantially both between individuals and from day to day within the same individual. Recent wearable-light data from a multi-country field study involving 191 adults and 1,480 participant-days demonstrated substantial variation in personal melanopic

exposure across individuals, activities, environments, sites, and days³. These findings suggest that real-world daytime melanopic exposure yields a highly variable light-exposure history from day to day within the same individual. However, it remains unknown whether naturally occurring day-to-day variation in daytime melanopic exposure within the same individual is associated with subsequent evening melatonin suppression.

This study primarily aims to determine whether naturally occurring day-to-day variation in daytime melanopic exposure, particularly daytime mEDI dose, within the same individual is associated with subsequent evening melatonin suppression. To address this aim, the study combines continuous monitoring of DLE in everyday environments with repeated-laboratory-based assessments of light-induced melatonin suppression. As a secondary, hypothesis-generating aim, we examine whether between-person differences in mean daytime mEDI dose are associated with differences in melatonin suppression. We further explore whether within-person variation in daytime mEDI dose is associated with melanopsin-mediated pupil responses and whether within-person variation in additional characteristics of DLE is associated with either physiological outcome.

Based on these aims, we specify one confirmatory hypothesis (CH1) and three hypothesis-generating research questions:

CH1: Primary confirmatory hypothesis

On average across participants, evening melatonin suppression will be lower on bright-light session days with a higher wake-to-laboratory-arrival daytime mEDI dose relative to the participant's own average across bright-light session days.

Hypothesis-generating research questions (HG):

HG1: Are between-person differences in mean wake-to-laboratory-arrival daytime mEDI dose across bright-light session days associated with differences in average evening melatonin suppression?

HG2: Is within-person variation in wake-to-laboratory-arrival daytime mEDI dose across bright-light session days associated with melanopsin-mediated pupil response?

HG3: Is within-person variation in additional prespecified DLE characteristics (mEDI dose during daytime one and two days before the laboratory visit, mEDI dose during sleep, and 3 h before habitual bedtime; duration of exposure above 1,000 lux mEDI; midpoint of the brightest 10-h mEDI period) associated with melatonin suppression (HG3a) or pupil-related outcomes (HG3b)? Each exposure metric and outcome will be evaluated in a separate hypothesis-generating model.

By testing whether naturally occurring light history is associated with day-to-day variation in light responsiveness in the circadian system, this study may help the development of recommendations for healthy light exposure and interventions targeting personal light exposure behaviours.

Methods

Sampling Plan

Ethics and Informed consent

An ethics application to the TUM Ethics Committee is in preparation. The informed consent form will be made available as part of the online screening. Participants will be required to provide electronic consent before completing the online screening questionnaire. Eligible participants will be asked to review the consent form again and provide written informed consent during the in-person screening before study enrolment, in accordance with the Declaration of Helsinki.

Inclusion and Exclusion criteria

In this study, we will enrol up to 60 healthy young adults. Recruitment will cease when either 50 participants have contributed three valid paired observations for CH1 or 60 participants have been enrolled, whichever occurs first. Melatonin suppression depends on the amount of short-wavelength light reaching the retina. Age-related changes in the crystalline lens are known to reduce short-wavelength light transmittance and affect circadian physiological responses, such as melatonin suppression^{13–15}. To minimise the influence of such age-related optical changes on the assessment of melatonin suppression, the study population is restricted to participants aged 18 to 35 years. The detailed inclusion and exclusion criteria are summarised in Tables 1 and 2. These criteria were selected to minimise factors that may influence melatonin suppression and pupil-response outcomes.

Table 1. Inclusion criteria in this study.

Domain	Criterion	Assessment method	Time of assessment
Age	≥18, ≤35 years old	Self-report	Screening (online questionnaire)
Ability to follow instructions in English	Good English proficiency	Self-report	Screening (online questionnaire)
Physical and mental health	Good physical and mental health	Self-report	Screening (online questionnaire)

Ocular and retinal health	Good ocular and retinal health	Self-report	Screening (online questionnaire)
Visual acuity	Normal or corrected-to-normal acuity	Landolt C chart or similar	Screening (in-person)
Colour vision	Normal colour vision	HRR colour vision test	Screening (in-person)

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150 Table 2. Exclusion criteria in this study.

Domain	Criterion	Assessment method	Time of assessment
Body Mass Index (BMI)	Underweight or obese, <18, >30 kg/m ²	Measured height and weight	Screening (in-person)
Medication use	Any regular or habitual use of systemic medications, except antiallergic medication	Self-report	Screening (online questionnaire)
Sleep aids use	Any regular or habitual use (≥1/week) of sleep aids	Self-report	Screening (online questionnaire)
Health condition	History of diagnosis of any chronic, hormonal, ocular or neurological diseases or disorders	Self-report	Screening (online questionnaire)
Mental health	History of diagnosis of major depressive disorder, persistent depressive disorder, generalised anxiety disorder, personality disorder, post-traumatic stress	Self-report	Screening (online questionnaire)

	disorder, panic disorder, seasonal affective disorder, bipolar disorder, or an eating disorder.		
Vision correction	Refractive error ≤ -6.0 or $\geq +6.0$ diopters	Self-report	Screening (online questionnaire)
Nicotine consumption	≥ 2 times/week	Self-report	Screening (online questionnaire)
Recreational drug use	Any regular use of recreational drugs (≥ 2 times/week)	Self-report	Screening (online questionnaire)
Alcohol abuse	Excessive alcohol use	AUDIT ≥ 8	Screening (online questionnaire)
Sleep quality	Poor sleep quality	PSQI > 5	Screening (online questionnaire)
Sleep duration	Extremely short or extremely long sleep duration on workdays	Subjective sleep duration on workdays < 6 or > 10 hours (based on the MCTQ)	Screening (online questionnaire)
Sleep habit	Extreme late bedtime	Bedtime $> 1:30$	Screening (online questionnaire)
Daytime sleepiness	Excessive daytime sleepiness	ESS > 10	Screening (online questionnaire)
Night shift work	Night shift work in the past 1 month	Self-report	Screening (online questionnaire)
Time zone travel	Inter-time zone travel > 1 hour in the past one month	Self-report	Screening (online questionnaire)
Food-related allergies	Allergy or intolerance for food ingredients in experimental meals	Self-report	Screening (online questionnaire)

Hormone intake	Intake of any exogenous hormones (with exception of contraceptive hormones)	Modified RSQ	Screening (online questionnaire)
Hormonal and/or reproductive disorder	Diagnosis of hormonal and/or reproductive disorder	Modified RSQ	Screening (online questionnaire)
Current or past pregnancy	Being currently pregnant or having had a pregnancy or having been breast-feeding in the last 12 months	Modified RSQ	Screening (online questionnaire)

Abbreviations: BMI, Body Mass Index; AUDIT, Alcohol Use Disorders Identification Test¹⁶; PSQI, Pittsburgh Sleep Quality Index¹⁷; MCTQ, Munich Chronotype Questionnaire¹⁸; ESS, Epworth Sleepiness Scale¹⁹; RSQ, Reproductive Status Questionnaire²⁰ (Questions modified to cover those covered by other questionnaires were removed).

Recruitment and screening

Participants will be recruited through flyers distributed around Tübingen, social media, university mailing lists, and word of mouth. Interested individuals will first complete an online screening questionnaire implemented on REDCap^{21,22}, which contains questionnaires assessing participants' eligibility listed in Tables 1 and 2. Eligible individuals will then be invited to an in-person screening, where the remaining eligibility criteria will be tested. During in-person screening, participants will receive instructions for the study procedures and installation of the MyCap app²³ for at-home questionnaires during the study.

For the recruitment stopping rule, a complete participant will be defined as a participant who contributes three valid paired observations (melatonin suppression and light exposure) for CH1 after application of the prespecified quality-control criteria. Recruitment will continue until 50 complete participants have been obtained or 60 participants have been enrolled, whichever occurs first. Participants with two valid paired observations will remain eligible for CH1 because they contribute within-person information, whereas participants with fewer than two valid paired observations will not contribute to CH1. Recruitment decisions will be based solely on data completeness and the prespecified quality-control criteria, without examination of outcome values, estimated associations, or Bayes factors.

Sample size justification

Previous studies investigating the effects of prior light exposure on non-visual responses to light have typically used controlled laboratory manipulations and relatively small samples of between 8 and 24 participants. For example, Hébert et al. reported melatonin suppression of 53% versus 41% following dim and bright light history exposure, respectively¹⁰. Smith et al. observed mean melatonin suppression of $85.7\% \pm 6.5\%$ after a dim light history and $71.2\% \pm 7.1\%$ after a brighter light history¹¹. Chang et al. reported a mean difference in melatonin suppression of 35% (SD = 23%) between dim and typical indoor light (90 lux) history conditions¹². These studies demonstrate that prior light exposure can substantially affect circadian light responses under tightly controlled laboratory conditions. However, the present study examines naturally occurring DLE patterns under real-world conditions. Compared to laboratory manipulations, real-world light exposure is expected to exhibit greater variability and measurement noise, and therefore, smaller associations between predictors and outcomes are anticipated.

The primary objective of this study is to test whether within-person variation in naturally occurring DLE, specifically wake-to-laboratory-arrival daytime mEDI dose, is associated with variation in melatonin suppression in the evening. Confirmatory analyses will use Bayesian linear mixed-effects models with participant-specific random intercepts and slopes to account for repeated measurements within participants. To evaluate whether the planned sample size can provide compelling evidence for the confirmatory hypothesis, we conducted a fixed-sample Bayes Factor Design Analysis (BFDA)²⁴. The BFDA was conducted for CH1, which tests whether melatonin suppression is lower on the melatonin suppression measurement days when an individual's daytime mEDI dose is higher than their own average across the measurement days.

Because direct effect size estimates for associations between real-world DLE and melatonin suppression are not currently available, the outcome and predictor distributions were informed by relevant empirical datasets. Simulated melatonin suppression values were based on data obtained using a comparable experimental protocol, in which mean melatonin suppression was approximately 48.9 (SD = 27.7%; $n = 27$)²⁵. This distribution is also consistent with that reported by Chang et al. under comparable evening light exposure conditions (mean = 51.3%, SD = 25.2%)¹². The reference melatonin suppression data were subsequently standardised to a mean of zero and a standard deviation of one.

The predictor distribution, daytime mEDI dose, was informed by personal light exposure data from the German sample of the MeLiDos project²⁶. This dataset includes 58 participants, with a mean of 5.78 ± 0.77 measurement days per participant. Daytime melanopic-EDI dose showed a positively skewed distribution (median [IQR]: 3,440 [1,170–9,029] lux mEDI·h; mean $\pm$ SD: $6,895 \pm 10,029$ lux mEDI·h). Daytime mEDI dose was first transformed as $\log_{10}(\text{mEDI dose} + 0.1)$ ²⁷ and standardised. For each simulated

participant, one MeLiDos participant was sampled as a donor, and three daily exposure values were sampled from that participant's observed exposure profile. The sampled exposure values were then decomposed into a between-person component, defined as the participant-specific mean, and a within-person component, defined as each session-specific deviation from that mean. The between- and within-person components were standardised separately within each simulated dataset.

Simulated datasets included 50 participants, with three repeated observations of melatonin suppression and daytime mEDI dose per participant, and participant-specific random intercepts and slopes. Before conducting the BFDA, we checked whether the assumed data-generating model, including the specified random-effect and residual variability, produced plausible melatonin suppression data across different population mean intercepts from -0.9 to 0.9 at 0.3 step and population-level within-person slopes from -0.5 to 0.5 at 0.1 steps. Candidate random-intercept and random-slope standard deviations ranged from 0.1 to 0.3 , residual standard deviations ranged from 0.3 to 1.0 , and intercept-slope correlations were -0.5 , 0 , or 0.5 . For each parameter combination, 100 datasets were simulated. Reference intervals for the mean and standard deviation of melatonin suppression were obtained from 5,000 bootstrap samples of the reference dataset. A parameter combination was considered plausible when at least 80% of its simulated datasets had both a mean and a standard deviation within the corresponding 95% bootstrap reference intervals.

For the BFDA, the standard deviations of the participant-specific random intercepts and within-person slopes were both set to 0.2 , with an intercept-slope correlation of zero, and the residual standard deviation was set to 0.6 . The population-level between-person association was set to zero. The retained population mean intercept-within-person slope combinations were then used as the data-generating conditions in the BFDA. For each condition, Bayes factors were computed using Bayesian multilevel models fitted with the "brms" package²⁸. The full model included population-level effects for both the within-person and between-person mEDI components, together with participant-specific random intercepts and random slopes for the within-person effect, while retaining the between-person effect and the same random-effects structure in the null model. Thus, the model comparison specifically evaluated evidence for the population-level within-person association while accounting for between-person differences in mean exposure and allowing the within-person association to vary across participants under both models.

Cauchy priors centred at zero with a scale of $\sqrt{2}/4$ were assigned to both population-level mEDI coefficients. The intercept was assigned a normal prior with mean zero and standard deviation one; standard-deviation and residual parameters were assigned Student-t priors with three degrees of freedom and a scale of 0.5 ; and the random-effect correlation matrix was assigned an LKJ(2) prior.

Models were initially estimated using four chains and 4,000 iterations per chain, including 1,000 warm-up iterations. Models failing the prespecified sampling diagnostics were refitted once using 8,000 iterations per chain, including 2,000 warm-up iterations. A fit was considered diagnostically acceptable when it had no divergent transitions or maximum tree-depth exceedances, all $\hat{R}$ values were < 1.01 , bulk and tail effective sample sizes were each > 400 , and the minimum Bayesian fraction of missing information was > 0.30 .

Marginal likelihoods were estimated using bridge sampling with the “bridgesampling” package²⁹, and BF_{10} was calculated as the marginal likelihood of the full model divided by that of the null model. Bridge sampling was repeated twice for each model comparison, and the simulation was considered valid only when both models passed the sampling diagnostics, finite Bayes factors were obtained, and the repeated bridge-sampling estimates produced the same $BF_{10} \geq 10$ decision. Simulations not meeting these validity criteria were counted as unsuccessful in the overall evidential-sensitivity estimate. One hundred simulations were conducted per data-generating condition. We defined evidential support as $BF_{10} \geq 10$ together with a negative posterior mean for the population-level within-person slope. For conditions with a non-zero negative within-person slope, the true positive rate (TPR) was calculated as the proportion of all simulations satisfying these evidential and validity criteria. For the null condition, the corresponding proportion was interpreted as the false-positive rate (FPR).

Under the condition that 50 participants completed three valid paired observations, the simulations indicated that when no within-person association was present ($\beta = 0$), none of the simulations yielded strong evidence for the full model in the hypothesised negative direction under either population-intercept condition (FPR = 0%). For a small within-person effect of $\beta = -0.1$, the TPR ranged from 4–5%. For $\beta = -0.2$, the TPR increased to 56–62%, and for $\beta = -0.3$, it reached 93–94%. For larger effects of $\beta = -0.4$ and -0.5 , the TPR ranged from 99% to 100%. Results were similar across the two retained population-intercept values of -0.3 and 0 .

We additionally assessed recovery of the participant-specific random-slope standard deviation because each participant contributed only three repeated observations. Across all simulated data-generating conditions, the recovery-valid rate was 100%, the maximum absolute relative bias was 14.7%, and the minimum 95% credible-interval coverage was 95%. No fitted random-slope standard deviations collapsed to near zero, and the maximum median interval-width-to-generating-value ratio was 1.67. All simulated conditions met the assessment criteria, supporting retention of the participant-specific random slope in the primary model under the assumed data-generating conditions.

Furthermore, we conducted a recruitment-and-missingness sensitivity analysis incorporating the planned maximum enrolment of 60 participants. Three completeness scenarios were examined: under the high-completion scenario, 90%, 7%, and 3% of

participants were assumed to contribute three, two, and one valid CH1 observations, respectively; the corresponding proportions were 85%, 10%, and 5% under the expected scenario and 70%, 20%, and 10% under the stress scenario. Across the high-completion, expected, and stress scenarios, the TPR for $\beta = -0.3$ was 94–95%, while the FPR for $\beta = 0$ was 0% in all scenarios.

Overall, under the assumptions of the present data-generating model, the planned sample size of 50 participants with three repeated bright-light observations per participant provides limited evidential sensitivity for a within-person association of $\beta = -0.1$, moderate sensitivity for $\beta = -0.2$, high sensitivity for $\beta = -0.3$, and near-complete sensitivity for associations of $\beta \leq -0.4$. Evidential sensitivity for $\beta = -0.3$, which is a middle effect size, remained high when incomplete observations and the maximum enrolment of 60 participants were incorporated into the simulations.

Experimental design

Participant scheduling

Unlike laboratory light manipulations, real-world DLE can vary substantially from day to day within the same individual due to factors such as work schedules, daily activities, and weather conditions. Consequently, a single assessment may not adequately capture an individual's typical light exposure history. To reduce the influence of such day-specific variability, the present study adopts a repeated-measures design, combining continuous light monitoring with multiple laboratory assessments of melatonin suppression and pupillometry. Accordingly, participants will complete four experimental sessions, during which melatonin concentration and pupillometry will be assessed repeatedly. One of the first three experimental sessions will be conducted as the dark-control session and the timing of the session will be randomised. The dark-control session provides the reference melatonin profile used to calculate melatonin suppression for all three bright-light sessions. Participants will complete two experimental sessions per week over two consecutive weeks, with at least two days between sessions. The sessions in the second week will be scheduled on the same weekdays as the corresponding sessions in the first week. The first experimental session will be scheduled at least five days after the in-person screening. Because the three bright-light sessions will use an identical light-exposure protocol and carry-over effects are not expected, session order will not be included as a predictor in the confirmatory models.

Circadian stabilisation

Participants will be asked to follow a regular sleep/wake cycle schedule (16 hours wake, 8 hours sleep, with ± 30 minutes deviation allowed) from at least five days prior to the first experimental session until the last experimental session. This sleep/wake schedule will be self-selected during the in-person screening. Circadian stabilisation will begin at least five days before the first experimental session and continue until completion of the final

session. Adherence to this schedule will be monitored continuously using self-reported sleep and wake times, as well as a visual check of actimetry and light exposure prior to each experimental session by the experimenter. In case of doubt or unclear results, the experimenter will consult with an additional member of the team. On the morning after an experimental session, participants will be allowed to sleep one hour longer than their self-selected schedule.

In-laboratory experimental procedure

Participants will arrive at the laboratory seven hours prior to their habitual bedtime (HBT). After compliance with their self-selected sleep-wake schedule is verified, they will first undergo tetrahydrocannabinol (THC) and alcohol screening using a urine multidrug test (Drug-Screen Multi 5, nal von minden GmbH, Moers, Germany) and a breath analyser (Breathalyzer, ACE Instruments GmbH, Freilassing, Germany), respectively. Then, participants will consume a standardised meal before undergoing pupillometry measurements with the retinaWISE instrument³⁰ in a custom-made dark experimental tent. Light levels in the tent during stimulus delivery via this instrument will be kept to a minimum (<1 lux mEDI). Pupillometry measurements will end at the latest 4.5 hours before HBT. Four hours prior to HBT, the dim light exposure lasting 120 minutes will start. After 90 minutes of dim light exposure, participants' pupils will be dilated using eye drops (Mydriaticum Stulln®, Tropicamide 5 mg/1 ml, Pharma Stulln GmbH, Stulln, Germany). Half an hour later, i.e. at two hours prior to HBT, participants will wear a virtual reality (VR) headset to view either a bright light of ~90 lux mEDI (n=3 experimental sessions) or a dark stimulus of <0.01 lux mEDI (n=1 dark control session). At the end of these two hours (i.e. at HBT), participants will remove the headsets and the experimental session will be completed.

Salivary samples for melatonin analysis will be collected every 30 minutes throughout the experimental session, starting four hours prior to HBT. From four hours to two hours before HBT, the sampling schedule will be organised in 30-minute "blocks" as follows (see Figure 1 inset A). Each block will start with a salivary sample, followed by three brief neurobehavioural questionnaires. From two hours prior to HBT to HBT, the same structure of this 30-minute experimental block will be kept, with the only difference being that participants will wear the VR headset to be exposed to dark or bright light for 20 minutes (Figure 1, inset B1 and B2, respectively). A total of four 20-minute dark or bright light exposures will be achieved during these two hours. Throughout the evening, participants will have no access to screens or light sources other than those included in the experimental protocol.

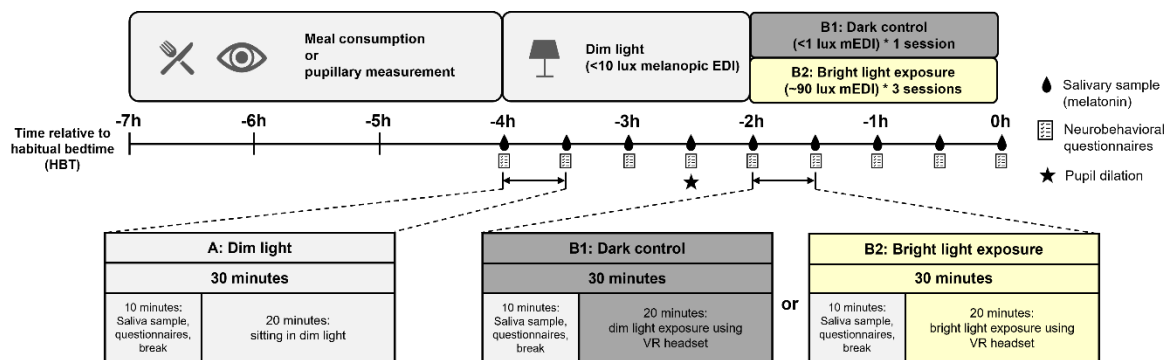


Figure 1. Experimental protocol for evening sessions. Participants will arrive in the laboratory 7 hours prior to HBT (habitual bedtime) and end the experiment at HBT. From 4 hours to 2 prior to HBT, the sampling schedule will be organized in 30-minute blocks, starting with the salivary samples, neurobehavioural questionnaires, a short break, followed by 20 minutes in dim light (inset A). The same sampling schedule will be kept during bright light exposure; however, here, the salivary samples, questionnaires, and breaks will be followed by a 20-minute delivery of the bright light stimulus via the VR headset (inset B2). Note that for the dark control session, participants will wear the VR headset but receive a light stimulus of intensity <0.01 lux mEDI (inset B1).

Light exposure stimuli

Light exposure will be delivered at the vertical plane using a VR head-mounted display (HTC Vive Pro Eye, HTC, Taiwan). This device comprises a binocular display with an in-built eye tracking technology (Tobii eye tracker, Sweden) for real-time pupillometry and eye-movement measurements. A detailed report on light stimuli delivery with this set-up and validation for melatonin suppression studies has been described by Fernandez-Alonso & Spitschan, 2025²⁵. The light stimulus for the control condition (dark) consists of a constant background, with intensity set to zero, yielding <0.01 lux mEDI (mean±SD across five VR headsets: 0.0076±0.0046 lux mEDI). The light stimulus for the bright light experimental sessions consists of a constant full-field polychromatic white light, generated by setting the red, green and blue channels to equal values at 50% intensity, yielding ~90 lux mEDI (mean±SD across five VR headsets: 90.94±0.28 lux mEDI). To calibrate the VR headsets, light intensities expressed in mEDI and photopic illuminance were measured for the left and right lenses of all five available VR head-mounted displays at gradually increasing intensity levels (from 0 to 1 at 0.1 steps), using a spectroradiometer (Jeti spectraval 1511-HiRes, Jena Technische Instrumente, Jena, Germany). This calibration procedure showed that a 50% input intensity corresponded to approximately 90 lux mEDI³¹. We selected this light level for use in our experiment carefully, seeking an optimum point in the dose-response curve for melatonin suppression. During the light exposure delivery, to ensure that participants keep their

eyes open and remain awake at all times, a visual stimulus (coloured dot, red, blue, green, and yellow, size set to 0.15 in arbitrary units) will be presented in the centre of the visual field, and participants will be prompted to press a button on a response pad every time it changes colour (random intervals set between 1 and 10 seconds). Responses will be recorded only to maintain wakefulness and will not be included in any analyses.

Pupillometry measurements

Pupillometry, including melanopsin-mediated pupil response (i.e. melanopsin sensitivity), will be performed with the retinaWISE binocular pupillometer³⁰ (Oculox Technologies, Muzzano, Switzerland). The device enables the delivery of silent substitution stimuli, which can change the activation of one or more photoreceptors while keeping other photoreceptors equally activated³². In the current study, three pupillometry paradigms will be used to assess 1) the post-illumination pupillary response (PIPR), and 2) LMS stimulation (silent substitution stimuli for joint modulation of L, M and S cones and constant activation of melanopsin), and 3) melanopsin stimulation (silent substitution stimuli for selective modulation of melanopsin and constant activation of L, M and S cones). The light stimuli will be delivered to the participants' right eye through a 10-30° field stop to achieve annular stimulation. For the PIPR protocol, six narrowband short-wavelength and six narrowband long-wavelength stimuli will be presented alternately. As part of the LMS-directed and melanopsin-directed sequences, the protocol will start with a 3-minute adaptation to the background spectrum to achieve light adaptation, followed by 16 pulses of LMS-directed and melanopsin-directed contrast, respectively. For all three protocols, the stimulus consists of a 3-second pulse of light windowed by a half cosine ramp (500ms on each side). Furthermore, the stimulus is accompanied by a 1-second pre-stimulus baseline and a 16-second post-stimulus measurement period.

The retinaWISE device will be placed in a black photography tent (Pop-Up Darkroom, ILFORD, Harman Technology Ltd., UK) to minimise external light exposure during the experiment. Participants will undergo a 10-minute dark adaptation period before completing the three pupillometry protocols (PIPR, LMS stimulation, and melanopsin stimulation). Throughout the experiment, participants will be asked to keep their gaze straight and limit head movements.

Salivary samples

Salivary samples for melatonin analysis will be collected every 30 minutes in each experimental session, starting 4 hours prior to participants' HBT, using Salivettes (Sarstedt, Nümbrecht, Germany). Samples (≥ 1 mL) will be centrifuged immediately after collection and stored at -20°C. The samples will be analysed for melatonin concentration using enzyme-linked immunoabsorbent assay (ELISA) by NovoLytiX GmbH, Switzerland, with typical specifications (limit of quantification 0.5–50 pg/mL, detection limit 0.5 pg/mL,

mean intra-assay precision: 7.8%, mean inter-assay precision 10.0%). In addition to melatonin concentration determination, one saliva sample will be collected at in-person screening session using an Oragene DNA collection kit (DNA Genotek Inc., Ontario, Canada) for genetic analyses. Genotyping will be performed for selected candidate polymorphisms previously implicated in melatonin suppression and pupil responses, including six PER2 single-nucleotide polymorphisms (SNPs) for haplotype inference³³, the PER3 variable-number tandem repeat (VNTR)³⁴, and the OPN4 I394T (rs1079610) polymorphism^{35,36}. PER2 will be classified according to Hap3 carrier status (carrier vs non-carrier) based on the 6 SNPs (rs4663302, rs147573126, rs190386281, rs142175638, rs11894535 and rs934945), PER3 according to the three VNTR genotypes (4/4, 4/5, and 5/5), and OPN4 according to two genotype groups (TT vs TC/CC), consistent with previous studies. The genotyping will be performed by Microsynth AG, Switzerland, with analysis based on PCR for OPN4 I394T and PER3 VNTR, and PCR and Sanger sequencing for six SNPs in PER2.

Personal light logging

To quantify DLE in the real-world, participants will continuously wear both a wrist-worn personal light logger (ActTrust 2, Condor Instruments, São Paulo, Brazil) and a chest-worn light logger (ActLumus, Condor Instruments, São Paulo, Brazil) from the in-person screening until completion of the final experimental session. The wrist-worn device with an acceleration sensor will be worn on the non-dominant wrist throughout the experimental period to assess compliance with the self-selected sleep/wake schedule. Continuous monitoring from the chest-worn device with recording epoch at 60-s intervals will be used to characterise participants' photic history and derive the melanopic light exposure metrics. Among these metrics, daytime mEDI dose on the day of each experimental session will be used as the primary predictor in the confirmatory analyses. Participants will be instructed to remove the light logger in case of showering, contact sports and/or swimming, and to record each removal and reattachment, including its start and end times and reason, in a simple wear and light-behaviour log³⁷. Because the chest-worn logger measures light exposure at the device location rather than at the eye, participants will also record any removal or change of spectacles, use of sunglasses, use of a brimmed hat or hood that shades the eyes, and any occasion on which the chest-worn sensor is covered by clothing, a bag, or another object. Start and end times will be recorded for each event, and the records will be reviewed with the participant at each laboratory visit. Participants requiring refractive correction will be asked to wear spectacles rather than contact lenses during waking light-exposure monitoring and to use the same pair whenever possible. The spectral transmittance of each pair of spectacles and sunglasses that participants intend to use during the monitoring period will be measured at the in-person screening session using a custom-built spectral transmittance measurement system. If an additional pair is used

unexpectedly, participants will be asked to bring it to the next laboratory visit for measurement. Data from the logger will subsequently be processed to derive the predefined DLE metrics, including daytime mEDI dose, described in the Outcome Derivation section.

Neurobehavioural tests

During each experimental block, participants will complete questionnaires assessing mood using the International Positive and Negative Affect questionnaire Short Form (I-PANAS-SF)³⁸, visual comfort, and subjective sleepiness using the Karolinska Sleepiness Scale (KSS)³⁹. These measures will be administered using tablets available in each participant's booth and will be collected for exploratory analyses only. They will not contribute to the confirmatory analyses in this study.

Additional self-report measures

Throughout the study, participants will complete a morning sleep diary (Consensus Sleep Diary, core version)⁴⁰ using the MyCap application integrated with REDCap. The morning sleep diary will additionally be used to monitor compliance with the prescribed sleep-wake schedule. These measures will not contribute to the confirmatory analyses in this study.

Medicinal, dietary, behavioural and exercise restrictions during the study

Throughout the study period, participants should abstain from melatonin supplements (any form), sleep aids or supplements. Participants will also be asked to completely avoid recreational drugs and nicotine for the whole study duration. Furthermore, on the day of an experimental session, participants will be asked to avoid intake of antihistamines and non-steroidal anti-inflammatory drugs (NSAIDs). Alcohol consumption will not be allowed in the three days prior to an experimental session. Caffeinated drink consumption will also be limited to a maximum of one cup to be consumed in the morning (cut-off time: noon). On the day of an experimental session, participants will be asked to avoid consumption of bananas, walnuts, sour cherries, citrus fruits, chocolate and any coloured candy or gum after 15:00, as these can interfere with the measurement of salivary melatonin. Furthermore, on days of scheduled experimental sessions, participants will be asked to avoid extreme physical activity, i.e., activity that is out of their routine, at least 4 hours before coming to the laboratory. On bright-light experimental session days, participants will be asked to refrain from wearing sunglasses or brimmed hats or hoods that shade the eyes from wake time until arrival at the laboratory. Ordinary prescription spectacles may be worn as usual. If sunglasses or eye-shading headwear are required for safety, medical, or other unavoidable reasons, participants will record the item used and the start and end times of use. Adherence to these instructions will be verbally checked on the day of the experimental session, prior to the start of the

experiment. These restrictions are intended to minimise non-experimental influences on melatonin secretion, melatonin suppression, pupil responses, and the correspondence between device-measured and ocular light exposure.

Outcome Derivation

Primary exposure metric

The source exposure metric will be daytime mEDI dose on the laboratory visit day (mEDI_dose_daytime_visitday). For the confirmatory analysis, the primary predictor will be the within-person component of this metric, derived after log transformation as described below. DLE will be continuously recorded using a wearable chest-worn light dosimeter (ActLumus) providing spectral irradiance measurements. The light exposure data will be analysed using LightLogR⁴¹. For epochs during which a measured pair of spectacles or sunglasses was worn, the spectral irradiance recorded by the chest-worn device will be multiplied wavelength by wavelength by the measured spectral transmittance of the corresponding eyewear before calculation of mEDI. When spectacles and sunglasses are worn together, the combined spectral transmittance of both optical filters will be applied. Melanopic equivalent daylight illuminance (mEDI) will be calculated for each measurement epoch from the unadjusted or eyewear-adjusted spectral irradiance, as appropriate. Because the shading produced by a brimmed hat or hood depends on factors such as head orientation, solar position, and headwear geometry, no numerical correction will be applied for headwear use. The resulting measures will therefore be interpreted as eyewear-adjusted estimates derived from light measured at the device location, rather than as direct measurements of retinal light exposure.

The mEDI time series will be aligned to clock time and combined with sleep/wake schedule records to determine wake and sleep periods for each day. The primary light exposure variable for confirmatory analyses will be daytime mEDI dose on each bright-light laboratory session day. This variable, mEDI_dose_daytime_visitday, will be calculated from wake time until laboratory arrival on the day of the experimental session and transformed as $\log_{10}(x + 0.1)$ prior to statistical analysis because dose-based mEDI metrics are expected to show positively skewed distributions according to a previous study that measured DLE in the various sites²⁶. The constant offset of 0.1 will ensure that the same predefined transformation can be applied consistently to all dose-based exposure metrics, including exploratory metrics that may contain zero values.

For each participant, the mean of the log-transformed daytime mEDI dose values across their valid bright-light session days will be calculated. The within-person component (log_mEDI_daytime_within) will be defined for each session as the log-transformed dose for that session minus the participant's mean log-transformed dose. The between-person component (log_mEDI_daytime_between) will be defined as the

participant's mean log-transformed dose across valid bright-light session days. The two components will be standardised separately before model estimation. The within-person component will be standardised across session-level observations, whereas the between-person component will be grand-mean centred and standardised using one value per participant. The population-level coefficient for log_mEDI_daytime_within will constitute the primary estimand, while log_mEDI_daytime_between will be included to separate the association of interest from between-person differences in average exposure.

Additional exposure metrics for exploratory analyses

In addition to the primary exposure metric (mEDI_dose_daytime_visitday), the variables for hypothesis-generating analyses will be obtained from DLE data. Additional daytime mEDI dose variables will be derived for days preceding the laboratory visit (from wake time to 3 h before habitual bedtime), denoted as mEDI_dose_daytime_lagN, where N indicates the number of days before the laboratory visit (lag1 = one day before, lag2 = two days before). Additional time-of-day-specific exposure variables will include mEDI dose during the 3 h preceding habitual bedtime (mEDI_dose_prebed) and mEDI dose during the sleep period (mEDI_dose_sleep). In addition to dose-based metrics, the duration of high-intensity light exposure will be calculated as the total time on the laboratory visit day with mEDI >1,000 lux mEDI (mEDI_TAT1000). In addition to intensity-based measures, DLE patterns will be characterised using non-parametric circadian rhythm analysis applied to the mEDI time series⁴². Dose-based exploratory metrics will be transformed as $\log_{10}(x + 0.1)$ before analysis, whereas duration and timing-based metrics will remain on their original scale. The midpoint of the brightest 10-h mEDI period (M10_midpoint) will also be calculated. Days that do not meet the predefined light-exposure quality-control criteria will not contribute to the derivation of the corresponding exposure metric. Each DLE metric will be separated into within- and between-person components in the same manner as mEDI_dose_daytime_visitday before the hypothesis-generating analysis will be performed.

These variables will be used as predictor variables in statistical models evaluating their association with melatonin suppression. For the analyses, mEDI_dose_daytime_visitday and mEDI_TAT1000 will be used on each experimental day. mEDI_dose_prebed and mEDI_dose_sleep will be derived from the pre-sleep and during-sleep period at night immediately before each experimental session, respectively. M10_midpoint will be calculated from the complete day preceding each experimental session.

Confirmatory outcome: Salivary melatonin suppression

The primary outcome for the confirmatory analyses will be percentage melatonin suppression during evening bright-light exposure. Salivary melatonin concentration in

missing samples will be estimated using interpolation via the piecewise cubic Hermite interpolating polynomial method, combined with forward and backward filling (function “pchip” from the package pracma⁴³, latest version available at data analysis). Interpolation will be applied according to pre-defined quality-control criteria described below. Melatonin suppression for a given participant and for each experimental bright light session will be expressed as percentage suppression relative to the dark experimental session:

$$\text{Melatonin suppression (\%)} = (1 - \text{AUC}_{\text{bright}} / \text{AUC}_{\text{dark}}) * 100$$

Where AUC_{bright} and AUC_{dark} represent the area under the curve (AUC) of the melatonin profile from two hours prior to HBT to HBT in the bright and dark experimental sessions, respectively. The AUC will be calculated using Simpson’s rule. The same AUC_{dark} obtained during the dark-control session will be used as the reference condition for all three bright-light experimental sessions. Accordingly, each participant may contribute up to three melatonin suppression values. Participants with unusable dark-control melatonin profiles, as defined by the prespecified melatonin data quality-control criteria below, will be excluded from the melatonin suppression analysis because a valid AUC_{dark} reference cannot be calculated.

Hypothesis-generating outcomes: Pupil-response measures

For hypothesis-generating analysis, we will calculate pupil-response measures including melanopsin-mediated pupil responses, melanopsin response amplitude, and LMS response amplitude. The melanopsin-mediated pupil response is the ratio of the melanopsin response amplitude to the LMS response amplitude. We will use pupil data from the melanopsin stimulation protocol and from the LMS stimulation protocol to identify changes in pupil size uniquely due to melanopsin and not to other photoreceptors. Data from the dark control session will not be used for this analysis. The pre-processing pipeline will be the same for data obtained by both protocols, and include resampling of raw data to evenly spaced time points and filtering the raw data to remove physiologically impossible pupil values below 1 mm and above 10 mm, as well as blinks. Furthermore, missing values will be estimated using linear interpolation. For each trial, pupil size will be normalised to the baseline pupil size, i.e. to the mean pupil diameter one second before stimulus onset. Furthermore, the timing of each trial will be aligned to stimulus onset. Afterwards, for each trial, pupil size will be converted to a percentage change over time from baseline using the following formula:

$$\text{Delta pupil (\%)} = [(\text{pupil diameter baseline} - \text{pupil diameter at time } t) / \text{pupil diameter at baseline}] * 100$$

where the numerator Pupil diameter is the size of the measured pupil (i.e. right pupil, mm) in a given trial. Trials where >50% of baseline pupil size data is unavailable and trials

where interpolated data >30% of total trial data will be excluded. Afterwards, all trials for a participant's given experimental session will be averaged. In order to quantify only melanopsin-related changes in pupil size, we will calculate melanopsin contribution to pupil responses as

$$\text{Melanopsin-mediated pupil response} = \frac{\text{Melanopsin response amplitude}}{\text{LMS response amplitude}}$$

Where Melanopsin response amplitude is the area under the curve during stimulus presentation in the melanopsin stimulation protocol, and LMS response amplitude is the area under the curve during stimulus presentation for the LMS stimulation protocol. In addition, both melanopsin response amplitude and LMS response amplitude will be included as outcome variables in hypothesis-generating analysis.

Quality Control

General principles

All quality-control and exclusion criteria will be applied before any confirmatory or exploratory statistical analyses and before inspection of associations between DLE and melatonin suppression/pupil-response outcomes. Exclusion decisions will be based on pre-defined criteria related to protocol adherence, data completeness, and measurement quality. Exclusions will be documented together with the reason for exclusion. Unless otherwise stated, valid sessions from a participant will be retained even if other sessions from the same participant are excluded.

Participant-level exclusion

Participants will be excluded from all confirmatory and exploratory analyses if one or more of the following participant-level exclusion criteria are met:

1. Withdrawal of consent.
2. Identification after enrolment of a predefined exclusion criterion that invalidates participation in this study.
3. Inability to establish and maintain the predefined regular sleep-wake schedule during the circadian-stabilisation period, such that the participant cannot proceed to the experimental sessions.

Outcome-specific participant exclusion will be applied when a participant cannot contribute valid data to a given outcome under one or more of the following conditions:

1. **Melatonin suppression analysis:** Participants will be excluded if the dark-control melatonin profile is unusable because melatonin suppression cannot be calculated without a valid dark-control AUC reference. Participants who complete only the dark-control session will also not contribute to the melatonin suppression analyses because no suppression outcome can be calculated. In

addition, participants must contribute at least two valid melatonin suppression observations to analyses estimating within-person associations, as specified in the Statistical analysis section. These participants may nevertheless contribute to pupillometry analyses if valid pupillometry data are available.

2. **Pupillometry analysis:** Participants will be excluded only if no valid pupillometry session remains after trial- and session-level quality control. For analyses estimating within-person associations, participants must contribute at least two valid observations for the relevant pupillometry outcome.

Participants with incomplete session attendance will not be excluded solely because one or more experimental sessions are missing. Instead, all valid sessions will be retained for the relevant outcome-specific analyses.

Session-level exclusion

At the session level, only the affected session is considered invalid if one or more of the following predefined quality-control exclusion criteria are met:

1. The session is terminated before completion of the pupillometry assessment, such that neither valid pupillometry nor melatonin data can be obtained.
2. The participant does not meet the predefined session eligibility requirements before data collection, such as a positive alcohol or drug screening result or non-adherence to the sleep-wake schedule.
3. A major procedural or environmental failure affects the entire session, such as prolonged power failure, incorrect identification, or loss of all session records.

When a session is excluded, all remaining valid sessions from the same participant will be retained whenever possible and analysed.

If a session assigned to the dark-control condition is terminated or judged invalid during data collection or before the next scheduled session, the next scheduled experimental session will be conducted as the replacement dark-control session. No additional session will be scheduled beyond the four planned sessions. Consequently, the replacement will be possible only when at least one scheduled session remains.

If a dark-control session appears valid during data collection but is subsequently found to be unusable after completing all experimental sessions and the melatonin assay, no replacement or additional session will be scheduled. The participant will be excluded from the melatonin suppression analyses because no valid dark-control AUC reference is available but may remain eligible for pupillometry analyses if valid pupillometry data are available.

Light exposure data quality control

Light exposure data will be considered valid only if sufficient recordings are available to reliably calculate the predefined light exposure metrics. Non-wear periods and periods of sensor covering will be identified based on participant-reported wear logs collected

throughout the study. Epochs during which the chest-worn sensor was covered will be treated as invalid exposure recordings. For participants requiring refractive correction, epochs for which spectacle or sunglass wear cannot be determined will be treated as invalid for the calculation of eyewear-adjusted exposure metrics. Use of a brimmed hat or hood will not itself invalidate an epoch when the sensor remains uncovered. Specifically, at least 80% of the expected recordings within each predefined window must be available. Note that the predefined window for the primary exposure, mEDI_dose_daytime_visitday, is from the participant-reported wake time to the recorded laboratory arrival time on the corresponding session. Completeness will be calculated as the number of valid recorded epochs divided by the number of expected epochs within this window. Data will be considered invalid if one or more of the following predefined quality-control exclusion criteria are met:

1. More than 20% of recordings are missing or invalid within the predefined analysis window because of non-wear, sensor covering, technical malfunction, unavailable eyewear information, or use of unmeasured eyewear for which spectral transmittance could not subsequently be obtained.
2. Clearly erroneous or implausible recordings are identified, including prolonged periods of unchanged values potentially indicating sensor malfunction, prolonged periods of near-zero values during reported daytime wakefulness potentially indicating sensor obstruction, or timestamp errors such as duplicate or non-chronological timestamps or recordings outside the documented device-deployment period. Signal patterns alone will not automatically result in exclusion because stable or very low light exposure may reflect genuine environmental conditions.

Potentially erroneous or implausible recordings will be adjudicated before inspection of melatonin suppression, pupillometry, or exposure-outcome association results. Adjudication will be based on the raw light recordings, participant-reported wear and sensor-covering logs, sleep-wake records, and documented device-deployment times. Only periods determined to represent non-wear, sensor covering, or technical error will be classified as invalid, and the decision and reason will be documented.

Light exposure data that meet the predefined quality-control exclusion criteria will not be used to derive the corresponding daytime mEDI dose. Only the affected observation (session) will be excluded from analyses requiring the corresponding daytime mEDI dose, while all other valid observations from the same participant will be retained.

Melatonin data quality control

A melatonin profile or the melatonin suppression value from a given session will be considered invalid if one or more of the following predefined quality-control exclusion criteria are met:

1. Two or more saliva samples are missing within the AUC calculation window from HBT-2h to HBT because of unsuccessful sample collection, sample loss, contamination, or assay failure.
2. Clearly erroneous or implausible melatonin measurements are identified, such as values outside the assay's quantifiable range.
3. For the dark-control session, no two consecutive samples collected by HBT have melatonin concentrations at or above 3 pg/mL. In this case, the dark-control profile will be considered invalid as a reference profile.
4. Melatonin suppression cannot be calculated because the dark-control session or the corresponding bright-light session is excluded according to the predefined session-level or melatonin quality-control criteria.

If one saliva sample is missing, the missing value will be interpolated according to the predefined outcome-derivation procedure. Specifically, the missing value will be estimated using interpolation via the piecewise cubic Hermite interpolating polynomial method, combined with forward and backward filling (function `pchip` from the package `pracma`⁴³, latest version available at data analysis).

Pupillometry data quality control

Pupillometry data will be evaluated at both the trial and session levels. Data will be considered invalid if one or more of the following predefined exclusion criteria are met:

1. More than 50% of the baseline pupil size data are unavailable within a trial.
2. Interpolated data exceed 30% of the total data points within a trial.
3. Fewer than 8 of 16 trials remain valid for either the melanopsin-directed or LMS-directed sequence after trial-level exclusion.
4. The melanopsin-mediated pupil response cannot be calculated because either the melanopsin or the LMS response amplitude is unavailable.

Trials meeting either of the first two criteria will be excluded individually. If either sequence does not retain the minimum number of valid trials, the corresponding session will not be used in analyses of the melanopsin-mediated pupil response.

Statistical Analysis Plan

Confirmatory analyses will be conducted in R⁴⁴ (latest version available at data analysis) using Bayesian linear mixed-effects models with random intercepts and slopes, fitted using the “`brms`”²⁸ and “`bridgesampling`” packages²⁹ (latest version available at data analysis). The R and package version used will be reported. Invalid observations will be excluded from the relevant outcome analysis, while other valid observations from the same participant will be retained. No imputation of missing derived outcome or exposure values will be performed. The prespecified interpolation of one missing saliva sample is part of melatonin outcome derivation and is described above. Because the models

estimate within-person associations, participants must contribute at least two valid observations for the outcome included in a given analysis.

Melatonin suppression will be reported descriptively as a percentage. The melanopsin-mediated pupil response ratio, melanopsin response amplitude, and LMS response amplitude will be reported as hypothesis-generating pupil outcomes. For Bayesian model estimation, each outcome will be standardised to a mean of zero and a standard deviation of one within its outcome-specific analysis dataset. The primary predictor, daytime mEDI dose on the laboratory-visit day (mEDI_dose_daytime_visitday), will be transformed as $\log_{10}(\text{mEDI_dose_daytime_visitday} + 0.1)^{27}$. Within each outcome-specific analysis dataset, the transformed predictor will then be decomposed into within- and between-person components, referred to as log_mEDI_daytime_within and log_mEDI_daytime_between, respectively. Let x_{ij} denote the transformed predictor of participant i on session j . The participant-specific mean, $x_{Bi} = \bar{x}_i$, will represent the between-person component, and the session-specific deviation from this mean, $x_{Wij} = x_{ij} - \bar{x}_i$, will represent the within-person component. The within-person component will be standardised across all eligible session-level observations. The between-person component will be grand-mean centred and standardised using one participant-specific mean per participant.

Confirmatory inference for melatonin suppression will not be based on p-values and will instead be based on Bayes factors obtained from Bayesian multilevel models fitted using “brms”²⁸. Marginal likelihoods will be estimated using bridge sampling with the “bridgesampling” package²⁹, and the Bayes factor will be calculated as the ratio of the marginal likelihood of the full model to that of the corresponding null model. The full and null confirmatory models will use the same outcome distribution, between-person exposure term, random-effects structure, and priors for all shared parameters. They will differ only in whether the population-level within-person exposure effect is included. The null model will omit this population-level effect while retaining the participant-specific random slope for the within-person exposure component. This comparison will therefore evaluate evidence for a population-level within-person association while allowing the participant-specific slopes to vary under both models.

Consistent with the priors used in the BFDA, Cauchy priors centred at zero with scale $r = \sqrt{(2)/4}$ will be assigned to the population-level within-person and between-person exposure coefficients. The intercept will be assigned a normal prior with mean zero and standard deviation one. The residual standard deviation and all random-effect standard deviations will be assigned Student-t priors with three degrees of freedom, location zero, and scale 0.5. The random-effect correlation matrix will be assigned an LKJ prior with shape parameter 2. The null model will omit the population-level within-person coefficient but will use the same priors as the full model for all shared parameters.

Bayes factors will be reported as BF_{10} , representing evidence for the full model relative to the corresponding null model. Values of $3 \leq \text{BF}_{10} < 10$, $10 \leq \text{BF}_{10} < 30$, and $\text{BF}_{10} \geq 30$

will be interpreted as substantial, strong, and very strong evidence for the full model, respectively. Values of $1/3 \leq BF_{10} < 3$ will be considered inconclusive. Values of $1/10 \leq BF_{10} < 1/3$, $1/30 \leq BF_{10} < 1/10$, and $BF_{10} < 1/30$ will be interpreted as substantial, strong, and very strong evidence for the null model, respectively. A confirmatory hypothesis will be considered supported only when $BF_{10} \geq 10$ and the posterior mean of the population-level within-person exposure coefficient is negative. Here, we report the full and null models for the confirmatory hypothesis according to the Wilkinson-Rogers notion⁴⁵.

Model adequacy will be assessed before interpretation of the results. Residual-versus-fitted plots will be inspected to assess the linearity of the predictor-outcome relationship and homoscedasticity, and normal quantile-quantile plots will be inspected to assess residual normality. Posterior predictive checks will additionally be used to evaluate correspondence between the observed and model-implied outcome distributions. Sampling will be considered adequate when all monitored parameters have $\hat{R}$ values no greater than 1.01, both bulk and tail effective sample sizes are at least 100 times the number of Markov chains⁴⁶, and there are no divergent transitions or maximum tree-depth exceedances. Observations will not be excluded solely on the basis of their residuals, influence, or consistency with the hypothesised association. The correlated random-intercept and random-slope model will constitute the prespecified primary model structure.

For all analyses including a participant-specific random slope, the same prespecified model-simplification procedure will be applied. If a model does not meet the prespecified sampling-diagnostic criteria, it will first be refitted using a longer sampling run with increased iterations and, where necessary, appropriately adjusted sampling-control parameters. If the correlated random-intercept and random-slope model still cannot be reliably estimated, it will be refitted without the random intercept-slope correlation. If the uncorrelated random-intercept and random-slope model also fails to meet the diagnostic criteria, a random-intercept-only model will be used. For confirmatory model comparisons, the full and corresponding null models will use the same random-effects structure, likelihoods, and priors for shared parameters. The final model structure and reason for any simplification will be reported.

The hypothesis-generating analyses, including all pupil-response analyses, will be estimation-focused and will not be evaluated using the confirmatory Bayes-factor threshold or treated as additional tests of CH1. Posterior estimates and 95% credible intervals will be reported and interpreted in terms of their direction, magnitude, uncertainty, and consistency across related exposure metrics and pupil outcomes. No individual association will be presented as confirmatory evidence, and the results will be used to generate hypotheses for future confirmatory studies.

Sensitivity analyses will examine whether the association estimated in the confirmatory melatonin suppression analysis remains robust after adjustment for prespecified participant- and session-level covariates, including age, sex, menstrual

phase, chronotype, and genotype. The within-person exposure deviation, participant-specific mean exposure, and random-effects structure will be retained in these models. For each sensitivity analysis, the corresponding full and null models will contain the same covariates and will differ only in the population-level within-person exposure effect. These analyses will not replace the primary confirmatory analysis. No confirmatory sensitivity analyses will be conducted for the pupil-response outcomes.

These analyses, including research questions, hypotheses, outcome measures, sampling plan, analysis model, and interpretation, are summarised in Table 3 as a Registered Report design table.

Confirmatory hypothesis analysis

CH1. On average across participants, evening melatonin suppression will be lower on bright-light session days with a higher wake-to-laboratory-arrival daytime mEDI dose relative to the participant's own average across bright-light session days.

Primary model (CH1)

CH1Full: Melatonin suppression ~

log_mEDI_daytime_within + log_mEDI_daytime_between +
(1 + log_mEDI_daytime_within | Participant ID)

CH1Null: Melatonin suppression ~

log_mEDI_daytime_between +
(1 + log_mEDI_daytime_within | Participant ID)

Hypothesis-generating analysis

HG1. Are between-person differences in mean wake-to-laboratory-arrival daytime mEDI dose across bright-light session days associated with differences in average evening melatonin suppression?

HG1Model: Melatonin suppression ~

log_mEDI_daytime_within + log_mEDI_daytime_between +
(1 + log_mEDI_daytime_within | Participant ID)

HG2. Is within-person variation in wake-to-laboratory-arrival daytime mEDI dose across bright-light session days associated with melanopsin-mediated pupil response?

HG2Model: Melanopsin-mediated pupil response ~

log_mEDI_daytime_within + log_mEDI_daytime_between +
(1 + log_mEDI_daytime_within | Participant ID)

HG3. Is within-person variation in additional prespecified DLE characteristics (mEDI dose during daytime one and two days before the laboratory visit, mEDI dose during sleep, and 3 h before habitual bedtime; duration of exposure above 1,000 lux mEDI; midpoint of the brightest 10-h mEDI period) associated with melatonin suppression (HG3a) or pupil-related outcomes (HG3b)? Each exposure metric and outcome will be evaluated in a separate hypothesis-generating model.

HG3aModel: Melatonin suppression ~

mEDI metric_within + mEDI metric_between +
(1 + mEDI metric_within | Participant ID)

HG3bModel: Pupil-response outcome

(melanopsin-mediated pupil response, melanopsin response amplitude or
LMS response amplitude) ~
mEDI metric_within + mEDI metric_between +
(1 + mEDI metric_within | Participant ID)

where mEDI metric refers to mEDI_dose_daytime_lagN [N = 1 or 2], mEDI_dose_prebed, mEDI_dose_sleep, mEDI_TAT1000, and M10_midpoint, each evaluated in separate exploratory models for generating hypotheses. Dose-based exploratory metrics will be transformed as $\log_{10}(x + 0.1)$ before analysis, whereas duration- and timing-based metrics will remain on their original scales.

Sensitivity analysis

Sensitivity analyses will be conducted for the confirmatory hypotheses only. First, a basic covariate-adjusted sensitivity model will include age, sex, and chronotype (MSFsc: Midpoint of Sleep in Free days, sleep-corrected calculated from MCTQ). Second, a genetic sensitivity model will include the pre-specified genotype variables PER2: Hap3 carrier or non-carrier; PER3: VNTR 4/4, 4/5 or 5/5; OPN4: TT vs TC/CC. Third, a female-participant sensitivity model will examine menstrual phase (follicular or luteal) obtained from RSQ among female participants only. Participants using hormonal contraceptives and sessions for which menstrual phase cannot be reliably determined because of irregular cycles or insufficient information will not be included in this sensitivity analysis.

Basic sensitivity analysis (SA1)

949 **SA1Full:** Melatonin suppression
950 ~ log_mEDI_daytime_within + log_mEDI_daytime_between +
951 age + sex + chronotype +
952 (1 + log_mEDI_daytime_within | Participant ID)

953 **SA1Null:** Melatonin suppression
954 ~ log_mEDI_daytime_between + age + sex + chronotype +
955 (1 + log_mEDI_daytime_within | Participant ID)

956

957 **Genetic sensitivity analysis (SA2)**

958 **SA2Full:** Melatonin suppression ~
959 log_mEDI_daytime_within + log_mEDI_daytime_between +
960 PER2 + PER3 + OPN4 +
961 (1 + log_mEDI_daytime_within | Participant ID)

962 **SA2Null:** Melatonin suppression ~
963 log_mEDI_daytime_between +
964 PER2 + PER3 + OPN4 +
965 (1 + log_mEDI_daytime_within | Participant ID)

966

967 **Female-participant sensitivity analysis (SA3)**

968 **SA3Full:** Melatonin suppression ~
969 log_mEDI_daytime_within + log_mEDI_daytime_between +
970 age + chronotype + menstrual phase +
971 (1 + log_mEDI_daytime_within | Participant ID)

972 **SA3Null:** Melatonin suppression ~
973 log_mEDI_daytime_between +
974 age + chronotype + menstrual phase +
975 (1 + log_mEDI_daytime_within | Participant ID)

976

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978 **Study timeline**

979 Data collection is expected to begin following in-principle acceptance and to be
980 completed within approximately 10 months. Laboratory assays, data processing, and
981 statistical analyses are expected to require an additional 3 months. The Stage 2
982 manuscript is expected to be submitted by March 2028.

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985 **Code availability**

986 The analysis code for the sample size justification can be found on
987 https://github.com/tscnlab/EtoEtAl_PCIRRStage1_2026.

Data availability

All datasets generated in this study will be made publicly available online upon publication of the Stage 2 manuscript.

Author contributions

TE and MS conceptualised this study. TE, CG, MFA, JZ, and MS contributed to the study design and methodology. TE, JZ and MS developed the statistical analysis plan, and TE conducted the sample-size simulation. TE wrote the original draft and the manuscript. All authors reviewed and edited the manuscript. MS supervised the project and TE and MS acquired funding.

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Declaration of AI-assisted technologies

During the preparation of this manuscript, the authors used ChatGPT (OpenAI) to support English-language editing, the drafting and revision of manuscript text, the development and debugging of R code, and the clarification and evaluation of methodological and statistical approaches. All AI-assisted outputs were critically reviewed, verified, and revised by the authors. The authors take full responsibility for the content of the manuscript, the analysis plan, and the accompanying code.

Competing interests

The authors declare no competing interests.

1022 Table 3. Study design table.

Question	Hypothesis	Sampling plan	Analysis Plan	Rationale for deciding the sensitivity of the test for confirming or disconfirming the hypothesis	Interpretation given different outcomes	Theory that could be shown wrong by the outcomes
Within participants, is daytime mEDI dose on the bright-light session day associated with evening melatonin suppression?	Confirmatory hypothesis (CH1) On bright-light session days when a participant's daytime mEDI dose is higher than their own average across bright-light session days, evening melatonin suppression will be lower.	N = 50 healthy young adults (18–35 y), four sessions: one dark-control session and three bright-light sessions. CH1 therefore provides up to three suppression observations per participant.	Confirmatory hypothesis model CH1Full: melatonin suppression ~ log_mEDI_daytime_within + log_mEDI_daytime_between + (1 + log_mEDI_daytime_within Participant ID) CH1Null: melatonin suppression ~ log_mEDI_daytime_between + (1 + log_mEDI_daytime_within Participant ID) Model will be fitted using brms, and BF_{10} will be estimated using bridge sampling.	Because no direct effect-size estimate is available for the within-person association under real-world light exposure, we used a BFDA informed by empirical light-exposure and melatonin suppression data to assess evidential sensitivity across a range of plausible effect sizes. BFDA indicates that the TPR was 4–5% for $\beta = -0.1$, 56–62% for $\beta = -0.2$, 93–94% for $\beta = -0.3$, and 99–100% for $\beta = -0.4$ to -0.5 ; the FPR was 0% when $\beta = 0$.	CH1 will be supported if $BF_{10} \geq 10$ and the posterior mean of the within-person daytime mEDI dose coefficient is negative. BF_{10} from 1/3 to <3 will be inconclusive; $BF_{10} < 1/3$ will favour the null model, with stronger categories reported as prespecified in the Statistical Analysis section.	Controlled laboratory studies have shown that prior light exposure can influence subsequent melatonin suppression. These findings suggest that light sensitivity may also vary from day to day with naturally occurring differences in daytime light exposure. Our study will test whether this proposed relationship is detectable within individuals under real-world light-exposure conditions. Evidence supporting the null model would challenge the generalisability of the laboratory findings to this setting.
Hypothesis-generating question (HG1) Are between-person differences in	—	Same recruited sample as CH1, using all valid bright-light sessions with valid exposure and melatonin suppression data.	Hypothesis-generating (HG1) model HG1Model: Melatonin suppression ~ log_mEDI_daytime_within + log_mEDI_daytime_between + (1 + log_mEDI_daytime_within Participant ID)	—	The direction, magnitude, and uncertainty of the between-person coefficient will be reported descriptively.	—

mean daytime mEDI dose across bright-light session days associated with differences in average evening melatonin suppression?					Results will be treated as Hypothesis-generating and will not affect the interpretation of CH1.	
Hypothesis-generating question (HG2) Is within-person variation in daytime mEDI dose across bright-light session days associated with melanopsin-mediated pupil response?	—	The bright-light sessions with valid mEDI and pupillometry data.	Hypothesis-generating (HG2) model HG2Model: Melanopsin-mediated pupil response ~ log_mEDI_daytime_within + log_mEDI_daytime_between + (1 + log_mEDI_daytime_within Participant ID)	—	The direction, magnitude, and uncertainty of the within-person association will be reported descriptively. Results will be treated as hypothesis-generating.	—
Hypothesis-generating question (HG3) Is within-person variation in additional prespecified DLE metrics associated with melatonin suppression	—	Same recruited sample as CH1, using all sessions with valid exposure and melatonin suppression data and pupil-response outcomes data.	Hypothesis-generating (HG3) model HG3aModel: Melatonin suppression ~ mEDI metric_within + mEDI metric_between + (1 + mEDI metric_within Participant ID) HG3bModel: pupil-response outcome (melanopsin-mediated pupil response, melanopsin response amplitude, or LMS response amplitude) ~ mEDI metric_within + mEDI metric_between +	—	The direction, magnitude, and uncertainty of the within-person association will be reported descriptively. Results will be treated as hypothesis-generating.	—

(HG3a) or pupil-response outcome (HG3b)?			(1 + mEDI metric_within Participant ID) mEDI metric = mEDI_dose_daytime_lag1, mEDI_dose_daytime_lag2, mEDI_dose_prebed, mEDI_dose_sleep, mEDI_TAT1000, or M10_midpoint. Dose metrics will be log10(x + 0.1)-transformed; duration and timing metrics will remain on their original scales.			
Question for sensitivity analyses (SA1,2,3) Are the CH1 results robust to adjustment for basic covariates (SA1), genotype (SA2), and menstrual phase (SA3)?	On bright-light session days when a participant's daytime mEDI dose is higher than their own average across bright-light session days, evening melatonin suppression will be lower, after adjustment for basic covariates, genotype, and menstrual phase.	SA1 and SA2 will include participants contributing valid CH1 data and the required covariates. SA3 will be restricted to female participants with valid CH1 and menstrual-phase data.	SA1Full: melatonin suppression ~ log_mEDI_daytime_within + log_mEDI_daytime_between + age + sex + chronotype + (1 + log_mEDI_daytime_within Participant ID) SA1Null: melatonin suppression ~ log_mEDI_daytime_between + age + sex + chronotype + (1 + log_mEDI_daytime_within Participant ID) SA2Full: melatonin suppression ~ log_mEDI_daytime_within + log_mEDI_daytime_between + PER2 + PER3 + OPN4 + (1 + log_mEDI_daytime_within Participant ID) SA2Null: melatonin suppression ~ log_mEDI_daytime_between + PER2 + PER3 + OPN4 + (1 + log_mEDI_daytime_within Participant ID) SA3Full: melatonin suppression ~ log_mEDI_daytime_within + log_mEDI_daytime_between + age + chronotype + menstrual phase + (1 + log_mEDI_daytime_within Participant ID)	No separate evidential-sensitivity analysis was conducted for these sensitivity analysis. Their sample sizes may be smaller than that of CH1, particularly for SA3; their results will therefore be used to assess robustness and will not override the primary analysis.	Robustness will be assessed by comparing the direction, magnitude, uncertainty, and Bayes-factor evidence for daytime mEDI dose with the primary CH1 results. Sensitivity analyses will not override the primary analysis.	—

			SA3Null:melatonin suppression ~ log_mEDI_daytime_between + age + chronotype + menstrual phase + (1 + log_mEDI_daytime_within Participant ID) Chronotype will be defined as sleep-corrected midsleep on free days (MSFsc). Genotypes will be coded as PER2 Hap3 carrier/non-carrier, PER3 VNTR 4/4, 4/5, or 5/5, and OPN4 TT versus TC/CC. SA3 will be restricted to female participants. The menstrual phase (follicular or luteal) will be obtained from RSQ. Participants using hormonal contraceptives and sessions for which menstrual phase cannot be reliably determined will not be included in this sensitivity analysis.			
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